

TIA

Touchless Invoice Agent

For TASC Outsourcing — UAE manpower supply, scaled.

Any client's timesheet, in any format, all the way to a validated tax invoice in their inbox
— with people stepping in only by exception.

Team Cyberkunju · HackArena 2.0 · 2026-06-28

The problem we're solving

Mariam, FinOps Lead at TASC invoices **750+ clients** every month for **10,500+ associates** placed across UAE/GCC.

Today	Why it hurts
Timesheets arrive in 7 different shapes — clean Excel, scanned PDFs, photos of handwritten registers, plain emails	Every channel a new exception
FinOps re-keys each row into a consolidated Excel	200+ rows × 10+ clients × 30 days
Bills hand-checked against per-client contract rules, then emailed	Days of cycle, no visibility for the client
MOHRE WPS file, VAT, TRN, sequential numbering all manual	Compliance risk on every cycle

Existing AP tools read invoices. **Nobody reads handwriting in three formats AND validates the contract behind it.**

Our approach — 4 agents, deterministic core

1. **Data Capture Agent** — Ingests from **4 channels**: portal · email (direct / cc-silent / watched-mailbox) · online form · image/PDF OCR
2. **Verification Agent** — Hungarian assignment (scipy `linear_sum_assignment`) maps named rows → employee IDs; the **cost matrix is visible** in the Why drawer
3. **Statutory Compliance Agent** — **10 BTP-style contract-bound rules**, deterministic decimal math, UAE Federal Decree-Law 33/2021 OT multipliers, WPS SIF generator
4. **Commercial Billing Agent** — UAE Tax Invoice (TRN, VAT, sequential no., SAC code), Ramco SRP-shaped consolidated Excel, idempotent dispatch

The LLM only touches **handwriting OCR** and **grounded chat**. Money, validation, rendering — all deterministic, auditable, replayable.

Reference architecture (matches the brief's diagram)

4 channels → ingestion+OCR → Smart Bot + SAP → BTP validation → dispatch

- `events` table: append-only audit spine, unique idempotency keys
- 14 gold cases, eval gate 13/13 PASS (F1 + ECE)
- Smart Bot + SAP block emits **consolidated Excel** (Ramco-shaped) + **WPS SIF** (real MOHRE format)
- Per-client config: jurisdiction (UAE/KSA/IN), VAT rate, rate card, max OT %, dispatch order + grouping, watched mailboxes

Stack & no-wrapper credentials

Layer	What we picked	Why it's not a wrapper
Extraction	GLM-OCR (open-weight, self-hosted via Modal vLLM) + openpyxl + pdfplumber	We control inference; swap to any OpenAI-compatible endpoint
Resolution	rapidfuzz + jellyfish phonetic + scipy Hungarian	Globally optimal assignment, audit-visible cost matrix
Validation	10 pure-function rules, Decimal money math	Per-contract config; rule_id surfaced on every failure
Invoice	Typst (Rust compiler, deterministic PDF)	FTA-compliant tax invoice template, audit hash footer
Dispatch	Rust + axum + sqlx (port 8001), idempotency-keyed	Real microservice, not a Python stub
Chat	OpenAI tool-calling, strict citations, swap-ready via OPENAI_BASE_URL	Answers refuse without DB evidence

The hard cases — judged by what we route, not what we autofill

Case	Format	What TIA does
1, 8	Email — name only / ambiguous name	Hungarian sees 3 candidate IDs for "Aisha Al Zaabi" , routes to HITL with cost matrix
4	Handwritten image	GLM-OCR (markdown + KIE fallback) → 3 rows extracted, confidence per field
11	Typed PDF	pdfplumber text-layer → 3 rows in 80ms
13	Email against a FIXED_SCOPE contract whose SOW is already COMPLETED	Rule R5 sow_hours_not_exceeded fires — exactly the mentor's "completed early" case
14	Excel with 50 OT hours on 22 working days	Rule R4 ot_within_contract_cap — 28% > contract's 20% — routes to HITL

Eval: 13/13 PASS. Macro F1 $\approx$ 0.68 across 7 fields. ECE 0.07 (well-calibrated).

Results — every brief success measure addressed

Brief target	TIA today
80%+ touchless (auto routing)	<code>GET /metrics/stp</code> — live counter on Finance dashboard
Within minutes start to finish	<code>GET /metrics/time-to-invoice</code> — sub-second on Excel, ~2s on handwriting
99%+ accuracy , low-conf routed to a person	<code>GET /metrics/accuracy</code> — eval F1 published; HITL drawer for the rest
Mock SAP / Ramco	<code>/consolidate/{client}/{period}.xlsx</code> (30 SAP-shaped columns) + <code>payroll_processed_by_sap</code> event
WPS bank file	<code>/payroll/sif/{client}/{period}.sif</code> — SCR + EDR records, MOHRE filename pattern
UAE tax invoice	Typst PDF with "Tax Invoice" header, supplier+customer TRN, sequential no., VAT line, SAC for India

17 commits, 67/67 pytest, CI 4 jobs green, eval 13/13 on every push.

Roadmap — what swaps in on day 2

1. **Real Ramco SRP connector** — `/consolidate` already emits Ramco-shaped columns; needs only the API mapping
2. **Peppol PINT AE XML emit** — Invoice schema is already Peppol-shaped (TRN, VAT, sequential, place of supply); add the XML transform when UAE mandate goes live (June 2026)
3. **AIDA WhatsApp bridge** — Plug TIA's `/qa` into TASC's existing AIDA bot; Navaneeth's bridge is already running in the repo (`workers/whatsapp/`)
4. **Local-model swap for chat** — Same `/qa` endpoint, point `OPENAI_BASE_URL` at a Modal-served vLLM. Zero code change.
5. **Postgres 18 + pg_trgm** — SQLite today; one env var flip away

Thank you. Demo screen → next.